

On practical asymptotic stabilizability of switched affine systems

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Abstract

In this paper, we report some new results on practical asymptotic stabilizability of switched systems consisting of affine subsystems. We first briefly review some practical asymptotic stabilizability notions and some results from our previous papers. Then we propose a new approach to estimate the region of attraction for switched affine systems. Based on this new approach, we present several new sufficient conditions for the practical asymptotic stabilizability and global practical asymptotic stabilizability of such systems. Finally, a computational approach to check the new sufficient conditions is proposed and it is applied to several numerical examples.

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1. Introduction

A switched system is a particular kind of hybrid system that consists of several subsystems and a switching law orchestrating the active subsystem at each time instant. Recently, there has been increasing interest in the stability analysis and design of such systems (see [4,10,11,14,20] for some survey).

There have been many results on the stability analysis problem, most of which utilizes common or multiple Lyapunov or Lyapunov-like functions (see [4,10,11,14,20] and the references therein). The stability design problem is more challenging than the analysis one. It not only requires one to determine whether a given switched system is stabilizable, but also demands explicit constructions of stabilizing switching laws for stabilizable switched systems. Available results on this problem usually utilize a single quadratic Lyapunov function or multiple Lyapunov-like functions to derive sufficient conditions for stabilizability (see, e.g., [3,5,7,23,24]). There have also been some necessary and sufficient conditions for certain classes of switched systems in the literature (see, e.g., [5,13,25]). It should be noted that many of the switching strategies proposed in the existing literature either will cause Zeno in switchings (i.e., infinite switchings in finite time) which is not desirable in implementation, or cannot guarantee that the switching sequences thus generated are nonZeno. Efforts have been made to construct nonZeno switching sequences

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by either incorporating dwell-time or hysteresis in switching laws (see, e.g., [18,24]). However, by introducing such modified switching laws, one often needs to trade the asymptotic stabilizability of the equilibrium for the asymptotic convergence to a neighborhood around the equilibrium.

Up to now, most of the existing literature on stability issues of switched systems assumes that all subsystems share a common equilibrium (typically the origin) and hence the stability of such systems is actually the stability of the common equilibrium. However, recently in our research [26–30], we have noted that, under appropriate switching laws, switched systems whose subsystems have different or no equilibria may still exhibit interesting behaviors similar to those of conventional stable or asymptotically stable systems near an equilibrium. Such behaviors are defined as practical stabilizability (local behavior) and practical asymptotic stabilizability (behavior in a larger region) in these papers. They are natural extensions of the conventional concepts of practical stability [8,9], finite time stability [16,17,22], and ultimate boundedness [1,6] for ordinary differential or difference equations, which are concerned with bringing the system trajectory to be within given bounds around a given point. Similar boundedness behavior has also been observed by other researchers. For example, in [12,15], Lin studied the ultimate boundedness for switched linear systems with uncertainties. Such systems may not have a common equilibrium. It is therefore only reasonable to stabilize them within a neighborhood of the origin. In [21], Su et al. studied the practical stability of a class of discrete-time switched systems and proposed a switching law to keep the state trajectories within a neighborhood of a given point. Practical stability of discrete-time switched systems was also studied by Picasso and Colaneri in [19]. Zhai and Michel in [32,33] introduced the notion of practical stability for a class of continuous-time switched systems. It is worth noting that the practical stability and stabilizability concepts in the aforementioned papers are quite different from the notions of conventional Lyapunov stability and stabilizability for hybrid and switched systems and can lead to interesting results in stability analysis, tracking, and reachability problems. For a survey of practical stabilizability and its relationship to the conventional stabilizability of switched systems, the reader is referred to our recent paper [31] for more information.

The results reported in [26–29] are mainly concerned with the practical stabilizability of switched systems, which describes the local behavior of the system trajectories around the origin. In many cases, we are interested not only in a system's local behavior but also in its behavior in a larger region around the origin. For example, we may want to know whether the system can exhibit “convergent behavior” similar to that of a conventional asymptotically stable system. As a result of such a research endeavor, in [30], we formally introduced the notion of practical asymptotic stabilizability and presented some preliminary sufficient conditions by using energy functions; moreover, we applied the sufficient conditions to switched affine systems and presented some methods for estimating the region of attraction. Here we note that similar convergent behavior of switched affine systems about the origin has also been independently explored by Bolzern and Spinelli in [2]. Their results include some sufficient conditions for the quadratic stabilizability of the origin. Such conditions are closely related to the sufficient conditions in [30]. The “quadratic stabilizability” defined in [2] is closely related to the concept of practical asymptotic stabilizability. Compared to [2] and many other literature results on conventional stabilizability of switched systems, the results in [30] offer two distinct contributions. First, [30] presented results on estimating the region of attraction. Second, [30] proposed guaranteed nonZeno switching laws that can practically stabilize the system trajectory to any prespecified neighborhood around the origin.

In this paper, we continue our studies on the practical asymptotic stabilizability of switched affine systems. For such systems, we mainly adopt quadratic energy functions. There are two main contributions in the paper. First, we propose a new approach to estimate the region of attraction for switched affine systems. Such an approach exploits the special structure of affine subsystem vector fields and can derive larger estimated regions of attraction than those reported in [30]. Based on the approach, we present several new sufficient conditions for the practical asymptotic stabilizability and global practical asymptotic stabilizability of such systems. The new sufficient conditions are more general and easier to check than those reported in [30]. Second, we propose a computational approach to check the sufficient conditions. Such an approach can be easily implemented in a digital computer. We apply the approach to several examples to demonstrate its effectiveness.

We organize the paper as follows. In Sections 2 and 3, we briefly review some notions and results from our previous papers (in particular, [30]) which are related to the practical asymptotic stabilizability of switched systems. In Section 4, we focus on switched affine systems and report new results on estimating the region of attraction and some new sufficient conditions. A computational approach to check the sufficient conditions and some examples are then given in Section 5. Section 6 concludes the paper.

2. Switched systems and practical stabilizability notions

First let us briefly review some notions from [30]. In the sequel, we use $\|\cdot\|$ to denote the 2-norm, $B(x, r)$ to denote the open ball $\{y \in \mathbb{R}^n : \|y - x\| < r\}$ and $B[x, r]$ the closed ball. We use S_r to denote the r -sphere around the origin, i.e., $S_r = \{x \in \mathbb{R}^n : \|x\| = r\}$. By a domain D around the origin, we mean an open connected subset of $\mathbb{R}^n$ containing the origin.

A switched system is a dynamic system consisting of subsystems

$$\dot{x} = f_i(x), \quad i \in I \triangleq \{1, 2, \dots, M\}. \quad (1)$$

In this paper, we are particularly interested in *switched affine systems* whose subsystems are of the form

$$\dot{x} = A_i x + b_i, \quad A_i \in \mathbb{R}^{n \times n}, \quad b_i \in \mathbb{R}^n, \quad i \in I. \quad (2)$$

For a switched system, the active subsystem at each time instant is orchestrated by a switching law, which will be formally defined below. Given an initial state $x(0)$ at $t_0 = 0$, the switching law generates a switching sequence $\sigma = ((t_0, i_0), (t_1, i_1), \dots, (t_k, i_k), \dots)$ ($t_0 < t_1 < \dots < t_k < \dots, i_k \in I$) which indicates that subsystem i_k is active in $[t_k, t_{k+1})$. For a switched system to be well-behaved, we only consider *nonZeno* sequences which switch at most a finite number of times in any finite time interval. Such sequences are also said to be *valid* switching sequences. Switching sequences are generated by switching laws defined below.

Definition 1 (*Switching Law*). For the switched system (1), a switching law $\mathcal{S}$ (over $[0, \infty)$) is defined as a mapping $\mathcal{S} : \mathbb{R}^n \rightarrow \Sigma_{[0, \infty)}$ which specifies a nonZeno switching sequence $\sigma \in \Sigma_{[0, \infty)}$ for every initial state $x(0) \in \mathbb{R}^n$. Here $\Sigma_{[0, \infty)} \triangleq \{\text{switching sequence } \sigma \text{ over } [0, \infty)\}$. $\square$

Remark 1. $\mathcal{S}$ over $[0, \infty)$ as defined above is said to be *valid* since the switching sequences it generates are nonZeno. In the sequel, we simply call it switching law $\mathcal{S}$. A switching law is often determined by some rules or algorithms, which describe how to generate a valid switching sequence for a given $x(0)$, rather than mathematical formulae. $\square$

Next let us review some practical stability notions (see [30,31] for more details). We mainly discuss the case of the origin and let the initial time be $t_0 = 0$.

Definition 2. Assume that a switching law $\mathcal{S}$ is specified for the switched system (1). Given an $\epsilon > 0$, the system is

- *ϵ -practically stable (about the origin) under $\mathcal{S}$* if there is a $\delta = \delta(\epsilon) > 0$ such that

$$x(0) \in B(0, \delta) \Rightarrow x(t) \in B(0, \epsilon), \quad \forall t \geq 0. \quad (3)$$

- *ϵ -practically asymptotically stable on a given domain D around the origin under $\mathcal{S}$* if it is ϵ -practically stable under $\mathcal{S}$, and for every $x(0) \in D$, there is a $T = T(x(0)) \geq 0$ such that

$$x(t) \in B(0, \epsilon), \quad \forall t \geq T. \quad \square \quad (4)$$

Remark 2. One difference between ϵ -practical stability and conventional stability is that, in Definition 2, ϵ is a prespecified constant which cannot be varied. Therefore ϵ -practical stability concerns the boundedness of the trajectory in the ϵ -neighborhood around the origin. $\square$

Remark 3. An ϵ -practically asymptotically stable switched system is both ϵ -practically stable and ϵ -practically attractive on D under $\mathcal{S}$ (attractiveness is defined in the sense of Eq. (4)). $\square$

Definition 3. The switched system (1) is

- *practically stabilizable (about the origin)* if for every $\epsilon > 0$, there is a switching law $\mathcal{S} = \mathcal{S}(\epsilon)$ such that the system is ϵ -practically stable under $\mathcal{S}$.
- *practically asymptotically stabilizable on a given domain D around the origin* if for every $\epsilon > 0$, there is a switching law $\mathcal{S} = \mathcal{S}(\epsilon)$ such that the system is ϵ -practically asymptotically stable on D under $\mathcal{S}$.
- *globally practically asymptotically stabilizable* if it is practically asymptotically stabilizable on $D = \mathbb{R}^n$. $\square$

Remark 4. In Definition 3, ϵ can be varied as opposed to the fixed ϵ in Definition 2. Such a definition provides us with the flexibility in design and trajectory tracking problems where the bound ϵ may vary depending on the specific task we face and the accuracy we require. In general, for a practically stabilizable system, the smaller the ϵ value is, the more demanding the design of the switching law is; usually a smaller ϵ demands a faster rate of switching. In cases where a system is constrained by an upper bound on the switching rate, it may well be possible that the system can be made ϵ_1 -practically stable but cannot be made ϵ_2 -practically stable for some $\epsilon_1 > \epsilon_2 > 0$. Such systems, albeit interesting, cannot be called practically stabilizable by our definition. $\square$

Remark 5. For a practically asymptotically stabilizable system, given an $\epsilon > 0$, we can find a switching law $\mathcal{S}$ that can bring the trajectory within $B(0, \epsilon)$. The trajectory resembles that of a conventional asymptotically stable system. However, for a practically asymptotically stabilizable switched system, in general by using valid switching laws we can only keep the trajectory within certain ϵ -bound; yet we may not be able to drive it asymptotically to 0 as $t \rightarrow \infty$ (even though ϵ can be chosen to be small). $\square$

3. Sufficient conditions based on energy functions

Methods using energy functions similar to the conventional Lyapunov function methods can be applied to determine the practical attractivity of the switched system (1). In the sequel, we call a continuously differentiable function $V : \mathbb{R}^n \rightarrow \mathbb{R}$ an *energy function* if it is positive definite, i.e., $V(0) = 0$ and $V(x) > 0, \forall x \neq 0$. The following are some results from [30].

Theorem 1. Assume that an energy function $V(x)$ is given for the switched system (1). Also assume that a $\rho > 0$ is given and the set $\Omega_\rho = \{x \in \mathbb{R}^n : V(x) \leq \rho\}$ is bounded. The switched system (1) is practically asymptotically stabilizable on any domain $D \subseteq \Omega_\rho$ around the origin if the following two conditions are satisfied:

(a) For every $x \in \Omega_\rho - \{0\}$,

$$\min_{i \in I} \frac{\partial V}{\partial x} f_i(x) < 0. \quad (5)$$

(b) $0 \in \text{Int}(C)$. Here C is the convex hull of the set $\{f_i(0) : i \in I\}$, i.e., $C = \text{conv}(\{f_i(0) : i \in I\}) = \{\sum_{i=1}^M \lambda_i f_i(0) : \lambda_i \geq 0, i \in I \text{ and } \sum_{i=1}^M \lambda_i = 1\}$ and $\text{Int}(C)$ is the interior of C . $\square$

In the proof of Theorem 1 in [30], we have actually constructed a valid switching law (i.e., no Zeno switching sequences will be generated) which can practically asymptotically stabilize the switched system (1).

Remark 6. If the condition (a) in Theorem 1 can be further strengthened to be $\min_{i \in I} \frac{\partial V}{\partial x} f_i(x) < 0$ for every $x \in \mathbb{R}^n - \{0\}$, then the switched system (1) is globally practically asymptotically stabilizable. $\square$

Remark 7. The condition (b) in Theorem 1 guarantees that the switched system (1) is practically stabilizable (see [26,29] for proof) and the condition (a) guarantees that the system can be made attractive in a larger region. Together, they guarantee the practical asymptotic stabilizability. For our interest in the switched affine system (2) in this paper, condition (b) can be restated as

$$0 \in \text{Int}(C) \quad \text{where } C = \text{conv}(\{b_i : i \in I\}). \quad (6) \quad \square$$

Remark 8. As in many literature results on conventional stabilizability of switched systems which utilize a single Lyapunov function, in Theorem 1 we use a single energy function $V(x)$ for all subsystems. Such a function has the property that it remains continuous under any switching law. Therefore methods based on the classic Lyapunov approach can be extended to the study of practical stabilizability of switched systems. $\square$

Remark 9. In [2], Bolzern and Spinelli also studied the behavior of the switched affine systems (2) about the origin which is a nonequilibrium point. Their results include the following sufficient conditions for the quadratic stabilizability of the origin: there exist $\alpha_i \in (0, 1), i \in I$ such that $\sum_{i=1}^M \alpha_i = 1$, $A_{eq} = \sum_{i=1}^M \alpha_i A_i$ is Hurwitz, and $b_{eq} = \sum_{i=1}^M \alpha_i b_i = 0$. It is not difficult to see that there is close relationship between the condition $b_{eq} = \sum_{i=1}^M \alpha_i b_i = 0$ and the condition (b) in Theorem 1, and between the condition “ $A_{eq} = \sum_{i=1}^M \alpha_i A_i$ is Hurwitz” and the condition (a) in Theorem 1. Also note that the “quadratic stabilizability” in [2] is closely related to the

“practical asymptotic stabilizability” in Definition 3. Strictly speaking, for a switched affine system satisfying the sufficient conditions proposed in [2], to actually realize quadratic stabilization, Zeno switchings need to be aroused (as indicated in [2]). In [2], an implementable nonZeno switching law was suggested, which can only guarantee that the trajectory be convergent to a specified neighborhood around the origin. This is similar to what we mean by ϵ -practical asymptotic stability (for a fixed ϵ value). $\square$

4. Estimation of region of attraction and new sufficient conditions

Having reviewed some relevant notions and results, we now focus on the switched affine system (2). Due to the special structure of such systems, further insights and results on practical asymptotic stabilizability can be obtained.

First, we note that the vector fields of the affine subsystems can be decomposed into two parts which correspond to the vector fields of the following two auxiliary switched systems. The first one is a switched linear system with a common equilibrium at the origin

$$\dot{x} = A_i x, \quad i \in I. \quad (7)$$

The second one is an integrator switched system

$$\dot{x} = b_i, \quad i \in I. \quad (8)$$

For switched affine systems, we mainly consider quadratic energy function $V(x) = x^T P x$ where $P = P^T \in \mathbb{R}^{n \times n}$ is a real positive definite matrix. If $P A_i + A_i^T P$ is negative definite for every $i \in I$, $V(x)$ is called a *common quadratic Lyapunov function (CQLF)* for the first auxiliary switched system (7); if $P A_i + A_i^T P$ is negative semidefinite for every $i \in I$, $V(x)$ is called a *weak CQLF* for the system (7).

As mentioned in Remark 7, the condition in (6) is sufficient for the practical stabilizability of the switched affine system (2). Furthermore, we claim that for the switched affine system (2), if (6) holds true, then (5) must also hold in some neighborhood around the origin and hence the practical attractivity is guaranteed. This implies that under (6), the switched affine system (2) must be practically asymptotically stabilizable on some domain around the origin. To show the validity of the above claim, we will in fact explore the answer to the following more general question.

Question 1. *How to find an estimate of the region of attraction (i.e., the domain on which (5) holds true) for the switched affine system (2) if we know that (6) holds true?* $\square$

To answer Question 1, we assume that we have chosen a quadratic energy function $V(x) = x^T P x$. For each $i \in I$, the derivative of $V(x)$ along the state trajectory with respect to the i th subsystem is

$$\frac{\partial V}{\partial x} f_i(x) = x^T (P A_i + A_i^T P) x + 2x^T P b_i. \quad (9)$$

We may estimate the region of attraction by a $B(0, \tilde{r})$ where $\tilde{r} > 0$ is chosen such that (5) holds for each $x \in B(0, \tilde{r})$. Such a $\tilde{r}$ in fact exists. To find such a $\tilde{r}$, we consider αx , $\alpha > 0$ for each x on the unit sphere S_1 . In the sequel in Lemma 2, we will show that for every $x \in S_1$, when α is small enough,

$$\min_{i \in I} \frac{\partial V}{\partial x} f_i(\alpha x) = \min_{i \in I} \{ \alpha^2 x^T (P A_i + A_i^T P) x + 2\alpha x^T P b_i \} < 0 \quad (10)$$

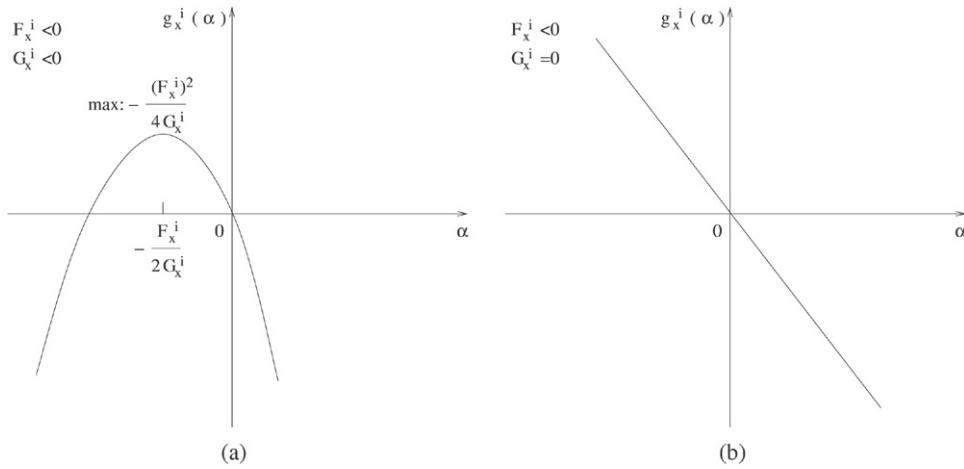
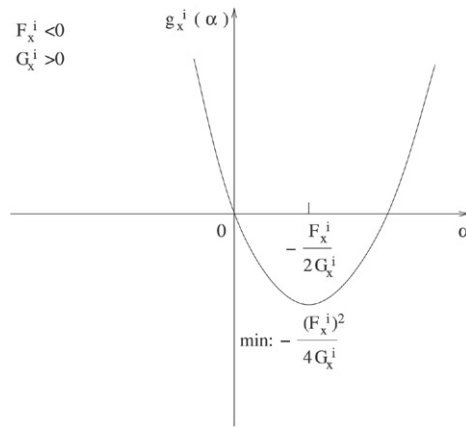
is always satisfied given (6). Hence for each $x \in S_1$, we can always find a maximum value $\alpha_{\max}$ such that $\min_{i \in I} \frac{\partial V}{\partial x} f_i(\alpha x) < 0$, $\forall 0 < \alpha < \alpha_{\max}$ (such an $\alpha_{\max}$ value will be regarded as ∞ if (10) holds for all $\alpha > 0$). Now by taking the minimum of all such $\alpha_{\max}$ values (corresponding to every $x \in S_1$), we can obtain $\tilde{r}$.

To formally carry out the aforementioned approach, we note that

$$\frac{\partial V}{\partial x} f_i(\alpha x) = \alpha^2 x^T (P A_i + A_i^T P) x + 2\alpha x^T P b_i \quad (11)$$

and for each $i \in I$ and for each $x \in S_1$ we define

$$F_x^i \triangleq 2x^T P b_i \quad (12)$$

Fig. 1. $g_x^i(\alpha)$ for Case 1, (a) when $G_x^i < 0$; (b) when $G_x^i = 0$.Fig. 2. $g_x^i(\alpha)$ for Case 2.

and

$$G_x^i \triangleq x^T(PA_i + A_i^T P)x. \quad (13)$$

Given an $i \in I$ and an $x \in S_1$, we note that (11) becomes a function of α only, which we define as

$$g_x^i(\alpha) \triangleq \frac{\partial V}{\partial x} f_i(\alpha x) = \alpha^2 G_x^i + \alpha F_x^i. \quad (14)$$

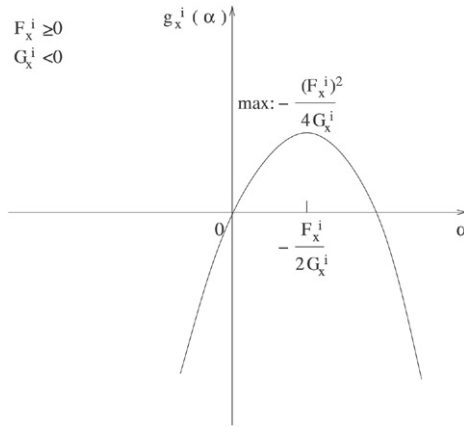
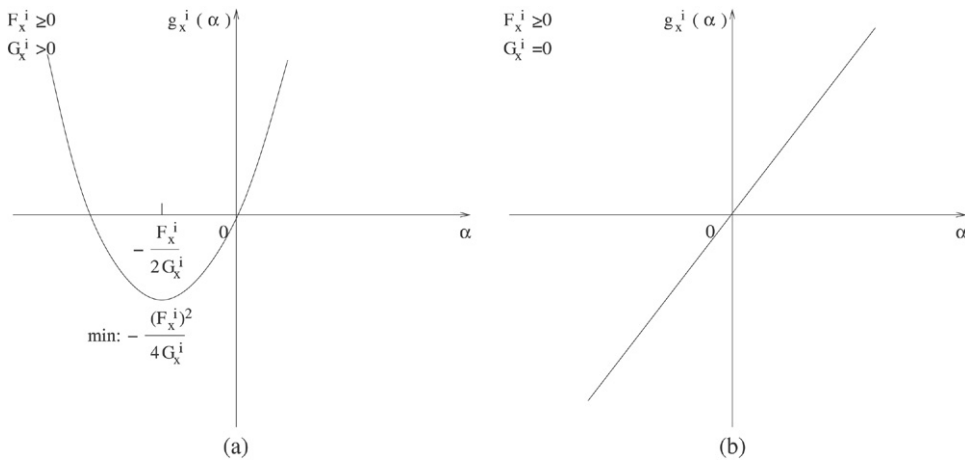
Given an $i \in I$, one of the following four cases must be true for an $x \in S_1$.

Case 1: $F_x^i < 0$ and $G_x^i \leq 0$

In this case, when $G_x^i < 0$, $g_x^i(\alpha)$ is a quadratic function with a maximum of $-\frac{(F_x^i)^2}{4G_x^i}$ (>0) at $\alpha = -\frac{F_x^i}{2G_x^i}$ (<0) (see Fig. 1(a)). When $G_x^i = 0$, $g_x^i(\alpha)$ is a linear function with slope F_x^i (<0) (see Fig. 1(b)). We define the set A_1^i as the set of $x \in S_1$ such that the condition of Case 1 is satisfied.

Case 2: $F_x^i < 0$ and $G_x^i > 0$

In this case, $g_x^i(\alpha)$ is a quadratic function with a minimum of $-\frac{(F_x^i)^2}{4G_x^i}$ (<0) at $\alpha = -\frac{F_x^i}{2G_x^i}$ (>0) (see Fig. 2). We define the set A_2^i as the set of $x \in S_1$ such that the condition of Case 2 is satisfied.

Fig. 3. $g_x^i(\alpha)$ for Case 3.Fig. 4. $g_x^i(\alpha)$ for Case 4, (a) when $G_x^i > 0$; (b) when $G_x^i = 0$.

Case 3: $F_x^i \geq 0$ and $G_x^i < 0$

In this case, $g_x^i(\alpha)$ is a quadratic function with a maximum of $-\frac{(F_x^i)^2}{4G_x^i} (\geq 0)$ at $\alpha = -\frac{F_x^i}{2G_x^i} (\geq 0)$ (see Fig. 3). We define the set Λ_3^i as the set of $x \in S_1$ such that the condition of Case 3 is satisfied.

Case 4: $F_x^i \geq 0$ and $G_x^i \geq 0$

In this case, when $G_x^i > 0$, $g_x^i(\alpha)$ is a quadratic function with a minimum of $-\frac{(F_x^i)^2}{4G_x^i} (\leq 0)$ at $\alpha = -\frac{F_x^i}{2G_x^i} (\leq 0)$ (see Fig. 4(a)). When $G_x^i = 0$, $g_x^i(\alpha)$ is a linear function with slope $F_x^i (\geq 0)$ (see Fig. 4(b)). We define the set Λ_4^i as the set of $x \in S_1$ such that the condition of Case 4 is satisfied.

Given an $x \in S_1$, we also decompose the subsystem index set I into I_x^1 , I_x^2 , I_x^3 , and I_x^4 . Here I_x^k contains all the subsystem index i such that $x \in \Lambda_k^i$, $k = 1, 2, 3, 4$.

Equipped with the above discussions and definitions, we introduce the following two lemmas.

Lemma 1. If (6) holds, then for every $x \in S_1$, there exists an $i \in I$ such that

$$F_x^i = 2x^T P b_i < 0. \quad (15)$$

Proof. We prove the lemma by contradiction. Assume that at some $x \in S_1$, $F_x^i = 2x^T P b_i \geq 0$ for all $i \in I$. Then $2x^T P y \geq 0$ for every $y \in C = \text{conv}(\{b_i : i \in I\})$. This leads to the conclusion that there exists a separating hyperplane between C and $\{0\}$, which is contradictory to the condition $0 \in \text{Int}(C)$. $\square$

Lemma 2. If (6) holds, then for every $x \in S_1$, when $\alpha > 0$ is small enough, (10) holds true (i.e., $\min_{i \in I} \frac{\partial V}{\partial x} f_i(\alpha x) = \min_{i \in I} \{\alpha^2 x^T (P A_i + A_i^T P) x + 2\alpha x^T P b_i\} < 0$).

Proof. Due to Lemma 1, for every $x \in S_1$, there exists an $i \in I$ such that $F_x^i < 0$. Since $g_x^i(\alpha) = \frac{\partial V}{\partial x} f_i(\alpha x)$ is a quadratic function in α , when $\alpha > 0$ is small enough, the linear term αF_x^i dominates and hence $g_x^i(\alpha) < 0$. Consequently (10) is established. $\square$

Now we present some new sufficient conditions based on the above case discussions.

Theorem 2. If (6) holds and

$$\bigcup_{i \in I} \Lambda_1^i = S_1, \quad (16)$$

then the switched affine system (2) is globally practically asymptotically stabilizable.

Proof. In view of (16), for every $x \in S_1$, there exists an $i \in I$ such that $x \in \Lambda_1^i$, i.e., $F_x^i < 0$ and $G_x^i \leq 0$. As can be seen from Case 1, for such an i , $g_x^i(0) = 0$ and $g_x^i(\alpha)$ is strictly decreasing for $\alpha > 0$. Hence $g_x^i(\alpha) < 0$, $\forall \alpha > 0$. Consequently, we conclude that (10) holds for every $x \in S_1$ and every $\alpha > 0$, which is equivalent to $\min_{i \in I} \frac{\partial V}{\partial x} f_i(x) < 0$, $\forall x \in \mathbb{R}^n - \{0\}$. In view of Remark 6, the switched affine system is globally practically asymptotically stabilizable. $\square$

Remark 10. In [30], we have shown that if (6) holds and if there exists a weak CQLF for the first auxiliary switched system (7), then the switched affine system (2) is globally practically asymptotically stabilizable. It should be noted that the condition in Theorem 2 is more general than the weak CQLF condition. In fact, if a weak CQLF exists for the system (7), then $G_x^i \leq 0$, $\forall x \in S_1, \forall i \in I$. This, combined with the result of Lemma 1, shows that for every $x \in S_1$, there exists some $i \in I$ such that $x \in \Lambda_1^i$. Hence (16) is satisfied. $\square$

If (16) is not satisfied, we have the following lemma.

Lemma 3. If (6) holds and

$$\bigcup_{i \in I} \Lambda_1^i \neq S_1, \quad (17)$$

then for every $x \in S_1 - \bigcup_{i \in I} \Lambda_1^i$, the set $I_x^2 \neq \emptyset$.

Proof. Consider an $x \in S_1 - \bigcup_{i \in I} \Lambda_1^i$. Due to Lemma 1, there exists an $i \in I$ such that $F_x^i < 0$. For such an i , since $x \notin \Lambda_1^i$, we must have $G_x^i > 0$. Consequently $x \in \Lambda_2^i$ and so $I_x^2 \neq \emptyset$. $\square$

The next theorem provides an estimate of the region of attraction.

Theorem 3. If (6) holds and $\bigcup_{i \in I} \Lambda_1^i \neq S_1$, then the switched affine system (2) is practically asymptotically stabilizable on any domain $D \subseteq \Omega_\rho = \{x \in \mathbb{R}^n : x^T P x \leq \rho\}$ around the origin, where $\rho < \min_{x \in S_f} V(x)$ and

$$\tilde{r} \triangleq \min_{x \in S_1 - \bigcup_{i \in I} \Lambda_1^i} \max_{i \in I_x^2} \left(-\frac{F_x^i}{G_x^i} \right). \quad (18)$$

Proof. We only need to consider those $x \in S_1 - \bigcup_{i \in I} \Lambda_1^i$. For such an x , by Lemma 3, $I_x^2 \neq \emptyset$. Hence there exists an $i \in I$ such that $F_x^i < 0$ and $G_x^i > 0$. In this case, α must be smaller than $-\frac{F_x^i}{G_x^i}$ to guarantee $g_x^i(\alpha) < 0$. Such a consideration, when applied to every $i \in I_x^2$ for every $x \in S_1 - \bigcup_{i \in I} \Lambda_1^i$, can then establish that the conditions of the theorem guarantee (5) for every $x \in B(0, \tilde{r})$. $\square$

Remark 11. In [30], a similar result to that of Theorem 3 is presented which says that the switched affine system (2) is practically asymptotically stabilizable on any domain $D \subseteq \Omega_\rho = \{x \in \mathbb{R}^n : x^T P x \leq \rho\}$ where $\rho < \min_{x \in S} \frac{\eta}{\lambda_{\max}} V(x)$; here the energy function $V(x) = x^T P x$ does not need to be a weak CQLF for the first auxiliary system (7) and $\eta = -\max_{x \in S_1} \min_{i \in I} 2x^T P b_i$ and $\lambda_{\max} = \max_{i \in I} \lambda_{\max, i}$ with $\lambda_{\max, i}$ being the maximum eigenvalue of $P A_i + A_i^T P$.

It is worth noting that the results in Theorems 2 and 3 are more general than the aforementioned result in [30]. For a $V(x) = x^T P x$ which is not necessarily a weak CQLF, there are two possibilities; either (16) is satisfied which establishes global practical asymptotic stabilizability, or (16) is not satisfied and then Theorem 3 provides an estimation of the region of attraction. In the second case, it must be true that $\tilde{r} \geq \frac{\eta}{\lambda_{\max}}$ since $\tilde{r}$ is the maximum radius value r such that for every $x \in B(0, r)$, there exists an $i \in I$ such that $2x^T P b_i < 0$ and $\frac{\partial V}{\partial x} f_i(x) = x^T (P A_i + A_i^T P)x + 2x^T P b_i < 0$. $\square$

Based on the results in Theorem 3, we have the following sufficient condition for the global practical asymptotic stabilizability of the switched affine system (2).

Theorem 4. Assume that (6) holds and $\bigcup_{i \in I} \Lambda_1^i \neq S_1$. If furthermore $I_x^3 \neq \emptyset$ for every $x \in S_1 - \bigcup_{i \in I} \Lambda_1^i$, then the switched affine system (2) is globally practically asymptotically stabilizable if

$$\hat{r} < \tilde{r} \quad (19)$$

where

$$\hat{r} \triangleq \max_{x \in S_1 - \bigcup_{i \in I} \Lambda_1^i} \min_{i \in I_x^3} \left(-\frac{F_x^i}{G_x^i} \right) \quad (20)$$

and $\tilde{r}$ is defined in (18).

Proof. For every $x \in S_1 - \bigcup_{i \in I} \Lambda_1^i$, since $I_x^3 \neq \emptyset$, there exists an $i \in I$ such that $F_x^i \geq 0$ and $G_x^i < 0$. In this case, α must be larger than $-\frac{F_x^i}{G_x^i}$ to guarantee $g_x^i(\alpha) < 0$. Such a consideration, when applied to every $i \in I_x^3$ for every $x \in S_1 - \bigcup_{i \in I} \Lambda_1^i$, can then establish (5) for every x satisfying $\|x\| > \hat{r}$. Also note that Theorem 3 guarantees (5) for every $x \in B(0, \tilde{r})$. Hence we conclude that, if $\hat{r} < \tilde{r}$, then (5) holds for every $x \in \mathbb{R}^n - \{0\}$. This leads to the global practical asymptotic stabilizability of the switched affine system (2). $\square$

Remark 12. In [30], we have shown that if $V(x) = x^T P x$ is not a weak CQLF but satisfies $\min_{i \in I} x^T (P A_i + A_i^T P)x \leq -\epsilon_0 \|x\|^2$, $\forall x \neq 0$, then $\min_{i \in I} \frac{\partial V}{\partial x} f_i(x) \leq \min_{i \in I} \{x^T (P A_i + A_i^T P)x\} + \zeta \|x\| \leq -\epsilon_0 \|x\|^2 + \zeta \|x\| < 0$ holds for all x satisfying $\|x\| > \frac{\zeta}{\epsilon_0}$, where $\zeta = \max_{i \in I} 2\|P b_i\|$.

It is worth noting that the results in Theorems 2 and 4 are more general than the aforementioned result in [30]. For a $V(x) = x^T P x$ which is not necessarily a weak CQLF and satisfies $\min_{i \in I} x^T (P A_i + A_i^T P)x \leq -\epsilon_0 \|x\|^2$, $\forall x \neq 0$, there are two possibilities; either (16) is satisfied which establishes the global practical asymptotic stabilizability, or (16) is not satisfied and then it must be true that $I_x^3 \neq \emptyset$ for every $x \in S_1 - \bigcup_{i \in I} \Lambda_1^i$. In the second case, it must be true that $\hat{r} \leq \frac{\zeta}{\epsilon_0}$ since $\hat{r}$ is the minimum radius value r such that for every x satisfying $\|x\| > r$, there exists an $i \in I$ such that $x^T (P A_i + A_i^T P)x < 0$ and $\frac{\partial V}{\partial x} f_i(x) = x^T (P A_i + A_i^T P)x + 2x^T P b_i < 0$. $\square$

Remark 13. For the convenience of obtaining a region of attraction, in this paper we simply choose P to be the identity matrix. In such a case, $B(0, \tilde{r})$ can simply be regarded as the region of attraction. However, we note that in Theorems 3 and 4, different choices of the matrix P will provide us with different $\tilde{r}$ and $\hat{r}$ values. How to choose the P matrix to obtain a better estimation is still an open problem which is under study. $\square$

5. A computational approach and examples

In this section, we propose a computational approach to check the conditions of the theorems in Section 4. The idea of the approach is as follows. In order to check the condition (16), we simply discretize the whole set S_1 into a sequence of states $x_1, x_2, \dots, x_K$ (K is chosen large so that the sequence is a good discretized version of S_1) and then

check if $I_x^1 \neq \emptyset$ for each $x_k, k = 1, 2, \dots, K$. If the answer is yes, then we can regard (16) as being satisfied. In the sequel, the same sequence of states will also be used to compute the approximate values of $\tilde{r}$ and $\hat{r}$.

Remark 14. The aforementioned discretization method is efficient and practical since both F_x^i and G_x^i are continuous with respect to x . A typical sequence of states in $\mathbb{R}^2$ can be chosen as $x_k = [\cos(k \frac{2\pi}{K}), \sin(k \frac{2\pi}{K})]^T, k = 1, \dots, K$. A typical sequence of states in $\mathbb{R}^3$ can be chosen as $x_k = [\sin(k_1 \frac{2\pi}{K_1}) \cos(k_2 \frac{2\pi}{K_2}), \sin(k_1 \frac{2\pi}{K_1}) \sin(k_2 \frac{2\pi}{K_2}), \cos(k_1 \frac{2\pi}{K_1})]^T$ for each $k = (k_1 - 1)K_2 + k_2$ where $1 \leq k_1 \leq K_1, 1 \leq k_2 \leq K_2$ (here $K = K_1 K_2$). $\square$

Once the sequence of states $\{x_k\}_{k=1}^\infty$ has been obtained, we construct two $M \times K$ matrices, $\tilde{A} = [\tilde{a}_{ik}]_{i,k=1}^{M,K}$ and $\hat{A} = [\hat{a}_{ik}]_{i,k=1}^{M,K}$ (here M is the number of subsystems for the switched affine system (2)) by the following algorithm.

Algorithm 1 (An Algorithm for Constructing $\tilde{A}$ and $\hat{A}$).

(1) Set $k = 1$.

(2) Consider x_k . For each $i \in I$, assign a value to $\tilde{a}_{ik}$ according to the rule

$$\tilde{a}_{ik} = \begin{cases} \infty, & \text{if } x_k \in \Lambda_1^i \\ -\frac{F_{x_k}^i}{G_{x_k}^i}, & \text{if } x_k \in \Lambda_2^i \\ 0, & \text{if } x_k \in \Lambda_3^i \cup \Lambda_4^i \end{cases} \quad (21)$$

and assign a value to $\hat{a}_{ik}$ according to the rule

$$\hat{a}_{ik} = \begin{cases} 0, & \text{if } x_k \in \Lambda_1^i \\ -\frac{F_{x_k}^i}{G_{x_k}^i}, & \text{if } x_k \in \Lambda_3^i \\ \infty, & \text{if } x_k \in \Lambda_2^i \cup \Lambda_4^i. \end{cases} \quad (22)$$

(3) If $k < K$, set $k = k + 1$ and go back to step (2). $\square$

In plain words, $\tilde{a}_{ik}$ corresponds to the maximum r value which guarantees $\frac{\partial V}{\partial x} f_i(\alpha x_k) < 0, \forall 0 < \alpha < r$, if subsystem i is active when the system state is $x = x_k$. $\hat{a}_{ik}$ corresponds to the minimum r value which guarantees $\frac{\partial V}{\partial x} f_i(\alpha x_k) < 0, \forall \alpha > r$, if subsystem i is active when the system state is $x = x_k$.

Once the matrices $\tilde{A}$ and $\hat{A}$ are computed, let

$$\tilde{\gamma} = \min_{1 \leq k \leq K} \max_{1 \leq i \leq M} \tilde{a}_{ik} \quad (23)$$

and

$$\hat{\gamma} = \max_{1 \leq k \leq K} \min_{1 \leq i \leq M} \hat{a}_{ik}. \quad (24)$$

The values of $\tilde{\gamma}$ and $\hat{\gamma}$ can then be used to come up with a unified approach to check the conditions of Theorems 2–4. The details are discussed below.

First of all, in view of the evaluation of $\tilde{a}_{ik}$ in (21), if

$$\tilde{\gamma} = \infty, \quad (25)$$

we conclude that $x_k \in \bigcup_{i \in I} \Lambda_1^i, \forall 1 \leq k \leq K$. If the discretization level is fine enough, we can then approximately regard that $\bigcup_{i \in I} \Lambda_1^i = S_1$ and consequently conclude that the system (2) is globally practically asymptotically stabilizable by applying Theorem 2.

Second, in view of (18) and (23), if the discretization level is fine enough, we can approximately regard

$$\tilde{r} \approx \tilde{\gamma}. \quad (26)$$

In view of (20) and (24), if the discretization level is fine enough, we can approximately regard

$$\hat{r} \approx \hat{\gamma}. \quad (27)$$

Once the values of $\tilde{r}$ and $\hat{r}$ have been obtained, Theorems 3 and 4 can then be applied.

Example 1. Consider a switched affine system (2) in $\mathbb{R}^2$ consisting of 3 subsystems $\dot{x} = A_i x + b_i$, $i = 1, 2, 3$, where

$$A_1 = \begin{bmatrix} 1 & 10 \\ -100 & 1 \end{bmatrix}, \quad b_1 = \begin{bmatrix} -0.7 \\ -0.7 \end{bmatrix}; \quad A_2 = \begin{bmatrix} -1 & 1 \\ -1 & -1 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 0.8 \\ -0.5 \end{bmatrix};$$

$$A_3 = \begin{bmatrix} -0.1 & 2 \\ -2 & -0.2 \end{bmatrix}, \quad b_3 = \begin{bmatrix} -0.5 \\ 0.8 \end{bmatrix}.$$

For this system, it can be verified that (6) holds true. Choose $V(x) = x^T x$. Choose $K = 2000$ and discretize S_1 into a sequence of states as mentioned in Remark 14. Algorithm 1 is applied to compute $\tilde{A}$ and $\hat{A}$. By (23) and (24), we have $\tilde{r} \approx \tilde{\gamma} = \infty$ and $\hat{r} \approx \hat{\gamma} = 0$. Hence, Theorem 2 can be applied to show that the switched affine system is globally practically asymptotically stabilizable. $\square$

Example 2. Consider a switched affine system (2) in $\mathbb{R}^2$ consisting of 3 subsystems $\dot{x} = A_i x + b_i$, $i = 1, 2, 3$, where

$$A_1 = \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \quad A_2 = \begin{bmatrix} -0.1 & 0.5 \\ -0.5 & 0.1 \end{bmatrix}, \quad b_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}; \quad A_3 = \begin{bmatrix} 0.5 & 0 \\ 0 & -0.2 \end{bmatrix},$$

$$b_3 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

For this system, it can be verified that (6) holds true. Choose $V(x) = x^T x$. Choose $K = 2000$ and discretize S_1 into a sequence of states as mentioned in Remark 14. Algorithm 1 is applied to compute $\tilde{A}$ and $\hat{A}$. By (23) and (24), we have $\tilde{r} \approx \tilde{\gamma} = 1.0329$ and $\hat{r} \approx \hat{\gamma} = \infty$. Hence, by Theorem 3, the switched affine system is practically asymptotically stabilizable on $B(0, 1.0329)$. This estimation of the region of attraction is better than $B(0, 0.8944)$ reported in [30]. $\square$

Example 3. Consider a switched affine system (2) in $\mathbb{R}^2$ consisting of 3 subsystems $\dot{x} = A_i x + b_i$, $i = 1, 2, 3$, where

$$A_1 = \begin{bmatrix} 0.1 & 0.2 \\ -0.01 & 0.1 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \quad A_2 = \begin{bmatrix} 0.1 & 0.015 \\ -0.5 & 0.1 \end{bmatrix}, \quad b_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix};$$

$$A_3 = \begin{bmatrix} 0.5 & 0 \\ 0 & -0.2 \end{bmatrix}, \quad b_3 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

For this system, it can be verified that (6) holds true. Choose $V(x) = x^T x$. Choose $K = 2000$ and discretize S_1 into a sequence of states as mentioned in Remark 14. Algorithm 1 is applied to compute $\tilde{A}$ and $\hat{A}$. By (23) and (24), we have $\tilde{r} \approx \tilde{\gamma} = 7.2524$ and $\hat{r} \approx \hat{\gamma} = 1$. Hence, by Theorem 4, since $\hat{r} < \tilde{r}$, the switched affine system is globally practically asymptotically stabilizable. For this example, in [30], we obtained $\frac{\eta}{\lambda_{\max}} = 2.2933$ and $\frac{\xi}{\epsilon_0} = 1.4142$. Note here $\tilde{r} > \frac{\eta}{\lambda_{\max}}$ as indicated in Remark 11 and $\hat{r} < \frac{\xi}{\epsilon_0}$ as indicated in Remark 12. This shows that the results in this paper are better than those in [30]. $\square$

Example 4. Consider a switched affine system (2) in $\mathbb{R}^3$ consisting of 4 subsystems $\dot{x} = A_i x + b_i$, $i = 1, 2, 3, 4$, where

$$A_1 = \begin{bmatrix} 0.5 & 0.5 & 0 \\ -0.5 & 0.5 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}; \quad A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.1 & 0.5 \\ 0 & -0.5 & 0.1 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix};$$

$$A_3 = \begin{bmatrix} 0.5 & 1 & 1 \\ -1 & 0 & -0.2 \\ -1 & 0.2 & -1 \end{bmatrix}, \quad b_3 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}; \quad A_4 = \begin{bmatrix} 0.5 & 0 & -0.2 \\ 0 & 1 & -0.1 \\ -0.1 & -0.2 & -2 \end{bmatrix}, \quad b_4 = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}.$$

For this system, it can be verified that (6) holds true. Choose $V(x) = x^T x$. Choose $K_1 = K_2 = 200$ and discretize S_1 into a sequence of states as mentioned in Remark 14. Algorithm 1 is applied to compute $\tilde{A}$ and $\hat{A}$. By (23) and (24), we have $\tilde{r} \approx \tilde{\gamma} = 0.4189$ and $\hat{r} \approx \hat{\gamma} = \infty$. Hence, by Theorem 3, the switched affine system is practically asymptotically stabilizable on $B(0, 0.4189)$. $\square$

6. Conclusion

This paper proposed a new approach to estimate the region of attraction for switched affine systems. Based on the estimation of the region of attraction, some new sufficient conditions for practical asymptotic stabilizability and global practical asymptotic stabilizability of such systems were proved. An efficient computational approach, which can be easily implemented in digital computers to check the sufficient conditions, was also developed. The results in the present paper are more general than those reported in our previous papers. A challenging and interesting future research topic is how to extend the results to switched systems with nonlinear subsystems.

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